Kevin Lee

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EDUCATION

University of California, Irvine, CA — **Sep 2025 – Dec 2026(Expected)**
Master's Degree, Embedded and Cyber-Physical Systems

National Taiwan University, Taipei, Taiwan — **Aug 2019 - Jul 2021**
Master's Degree, Mechanical Engineering

National Cheng Kung University, Tainan, Taiwan — **Aug 2014 - Jul 2018**
Bachelor's Degree, Mechanical Engineering

WORK EXPERIENCE

Hardware Engineer Summer Intern - Vehicle Embedded Engineer, WeRide Corp, San Jose — **Jun 2026 - Present**

- Developed and maintained drive-by-wire systems, supporting vehicle integration and debugging.
- Developed and deployed a nonlinear low-speed brake control, eliminating hunting and near-zero-speed torque overshoot.
- Diagnosed vehicle control issues using CAN logs and road tests, driving firmware fixes.
- Validated embedded control software through vehicle testing.

Senior Firmware Engineer, Delta Electronics Inc. — **Oct 2021 - May 2025**

- Developed **embedded firmware for EV traction inverters in C** as a production-release software component engineer.
- Performed CPU runtime load analysis and optimization, reducing system load from 55% to 35%.
- Automated the software release pipeline with python and dSPACE HiL, cutting verification effort by 85%.
- Led **ASPICE LV 2 & 3** project scope SW test as **Software Test Leader** w. certification awarded.
- Collaborated globally and provided technical support to customers and teams in the USA, India, Germany, Spain, and Italy.

Project Lead, Advanced Power R&D Center, National Taiwan University — **Aug 2019 - Jul 2021**

- Led a 6-person team to successfully complete two automotive cooperative research projects.
- Completed EV Truck by-wire HiL platform, subsystem development and powertrain integration.
- Designed and implemented 8 kW Plugged-In range extended hybrid power system for scooters.
- Built STM32-based VCU embedded software for energy-management control and CAN-FD communication.
- Built system energy consumption models with Simulink/dSPACE RTI for management strategy development.
- Planned and executed AVL dynamometer tests for WLTP and WMTC driving-cycle verification.
- Achieved 8.5% CO_2 reduction compared to a conventional ICE powertrain using the hybrid prototype.

MASTER'S THESIS RESEARCH

- Development of Energy Management Strategy for Range Extended Hybrid Scooter with HiL Validation and Well-to-Wheel CO2 Emissions Evaluation — **Aug 2019 - Jul 2021**

SKILLS

- Programming: C, C++, Python, Simulink, FreeRTOS, RTA-OS
- Development Platforms: Git, Jenkins, IBM Rhapsody, Enterprise Architect, IBM Rational,
- Embedded Platforms: STM32, ESP32, Infineon AURIX, Arduino, Raspberry Pi
- Validation Platforms: VectorCAST, HelixQAC, dSPACE, MicroAutoBOX, AVL Dyno
- Standards/Frameworks: MISRA C, ASPICE, ISO26262, HiL, CiL, MiL
- Communication Protocols: CANBUS, CANFD, I2C, SPI, UART